A Set-Theoretic Approach to Establishing Bijections between Integers and Rational Numbers

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Abstract: *This paper explores a novel set-theoretic approach to establishing a bijection between integers and rational numbers. We introduce two key concepts: a measure on collections of sets (μ) that maps to rational numbers, and a binary encoding rule (χ_i) that maps to integers. We analyze the potential for creating a bijection through this dual mapping and discuss challenges and possible solutions. We also explore the relationship between collections of sets and binary sequences, establishing a bijection between certain collections of sets and binary sequences, then demonstrating how to recover the original set collection from a given binary sequence.*

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1. Introduction

The countability of rational numbers is a well-established result in set theory. Standard proofs utilize techniques such as Cantor's diagonal argument, the "snake" enumeration, pairing functions, or applications of the Schröder-Bernstein theorem. This paper proposes an alternative approach using set- theoretic constructions and measure theory to establish a bijection between integers (represented as binary strings) and rational numbers. We also explore how binary sequences can be derived from collections of sets and vice versa, providing a foundation for our bijection approach.

2. Definitions and Notation

2.1 Set Collections

Let $Å = \{A_1, A_2, ..., A_n\}$ be a finite collection of finite sets, where each $A_i \subset \mathbb{N}$.

2.2 Set Union Ratio

Definition 2.1 (Set Union Ratio).

The Set Union Ratio $\mu: \mathcal{P}(\mathcal{P}(\mathbb{N})) \to [0,1]$ is defined as:

$\mu(Å) = |\cup_{A \in Å} A| / \sum_{A \in Å} |A|$

This measure represents the ratio between the cardinality of the union of all sets in the collection and the sum of individual set cardinalities.

Example 2.1 (Set Union Ratio).

Consider a collection $Å = \{\{1\}, \{1,2\}, \{1,2,3\}\}$:

$|\cup_{A \in Å} A| = |\{1,2,3\}| = 3$

$\sum_{A \in Å} |A| = 1 + 2 + 3 = 6$

Therefore, $\mu(Å) = 3/6 = 0.5$

2.3 Binary Encoding Rule

Definition 2.2 (Binary Encoding Rule).

The Binary Encoding Rule χ_i: $\mathcal{P}(\mathcal{P}(\mathbb{N})) \to \{0,1\}$ is defined for each position $i \in \mathbb{N}$ as:

$\chi_i(Å) = 1$ if $i = n + |A_n| - 1$ for some $A_n \in Å$
$\chi_i(Å) = 0$ otherwise

This generates a binary representation of the collection Å.

Equivalently, for a collection of sets $A = \{A_0, A_1, ..., A_{m-1}\}$, we define a binary sequence $\chi = (\chi_0, \chi_1, ..., \chi_{n-1})$ where:

- **$\chi_i = 1$ if i is one less than a cumulative sum of set cardinalities, i.e., $i \in \{|A_0|-1, |A_0|+|A_1|-1, |A_0|+|A_1|+|A_2|-1, ..., \Sigma_{j=0}^{k} |A_j|-1$ for $k=0,1,...,m-1\}$**
- **$\chi_i = 0$ otherwise**

This creates a mapping between collections of sets and binary sequences where the positions of 1s in the sequence correspond to the cumulative sums of set cardinalities (minus 1).

Example 2.2 (Binary Encoding).

Consider a collection $Å = \{\{1\}, \{1,2\}, \{1,2,3\}\}$:

For each set $A_n \in Å$:

Set A_0: $n=0$, $|A_0|=1$, so $i=0+1-1=0 \to \chi_0=1$

Set A_1: $n=1$, $|A_1|=2$, so $i=1+2-1=2 \to \chi_2=1$

Set A_2: $n=2$, $|A_2|=3$, so $i=2+3-1=4 \to \chi_4=1$

The binary encoding is: $\chi_0=1, \chi_1=0, \chi_2=1, \chi_3=0, \chi_4=1, \chi_5=0, ...$

Or written as a binary string: 10101...

Example 2.3 (Alternative View of Binary Encoding).

Consider a collection $A = \{A_0, A_1, A_2\}$ where $|A_0| = 1$, $|A_1| = 2$, $|A_2| = 3$. Using the cumulative sum approach with proper accounting for prior set indices:

Positions where $\chi_i = 1$:

For A_0: $i = 0 + |A_0| - 1 = 0 + 1 - 1 = 0$

For A_1: $i = 1 + |A_1| - 1 = 1 + 2 - 1 = 2$

For A_2: $i = 2 + |A_2| - 1 = 2 + 3 - 1 = 4$

Therefore, $\chi_0 = 1$, $\chi_2 = 1$, $\chi_4 = 1$, and all other positions are 0.

The resulting binary sequence is: 10101...

2.4 Binary Decoder and Recovery of Set Collections

Definition 2.3 (Binary Decoder).

The Binary Decoder $D: \{0,1\}^* \to \mathcal{P}(\mathcal{P}(\mathbb{N}))$ is defined as:

For a binary string $b = (b_0, b_1, b_2, ..., b_m)$, $D(b) = \mathring{A}$ where:
$\mathring{A} = \{A_n : b_i = 1 \text{ for } i = n + |A_n| - 1, 1 \leq n \leq k\}$

where each $A_n = \{1, 2, ..., |A_n|\}$ with $|A_n| = i - n + 1$ for each position i where $b_i = 1$.

Proposition 2.4 (Recovery of Set Cardinalities).

Given a binary sequence $\chi = (\chi_0, \chi_1, ..., \chi_{n-1})$, we can uniquely determine the cardinalities of the sets in the collection $A = \{A_0, A_1, ..., A_{m-1}\}$ that generated it according to Definition 2.2.

Let $I = \{i \mid \chi_i = 1\}$ be the set of indices where $\chi_i = 1$. Order these indices as $i_0 < i_1 < \ldots < i_{m-1}$.

By Definition 2.2, we know that:

$i_0 = |A_0| - 1$

$i_1 = |A_0| + |A_1| - 1$

$i_2 = |A_0| + |A_1| + |A_2| - 1$

...

$i_{m-1} = |A_0| + |A_1| + \ldots + |A_{m-1}| - 1$

From these equations, we can solve for the cardinalities:

$|A_0| = i_0 + 1$

$|A_1| = i_1 - i_0$

$|A_2| = i_2 - i_1$

...

$|A_{m-1}| = i_{m-1} - i_{m-2}$

Thus, we can recover the cardinalities of all sets in the collection.

□

Example 2.4 (Binary Decoding).

For the binary string b = 10101:

For each position i where $b_i = 1$:

$i = 0$: $b_0 = 1 \rightarrow$ Solve $0 = n + |A_n| - 1 \rightarrow n = 1$, $|A_1| = 1 \rightarrow A_1 = \{1\}$

$i = 2$: $b_2 = 1 \rightarrow$ Solve $2 = n + |A_n| - 1 \rightarrow n = 2$, $|A_2| = 2 \rightarrow A_2 = \{1,2\}$

$i = 4$: $b_4 = 1 \rightarrow$ Solve $4 = n + |A_n| - 1 \rightarrow n = 3$, $|A_3| = 3 \rightarrow A_3 = \{1,2,3\}$

Therefore, $D(10101) = \{\{1\}, \{1,2\}, \{1,2,3\}\}$

Example 2.5 (Recovering Set Cardinalities).

Consider the binary sequence $\chi = (1, 0, 1, 0, 1)$. The indices where $\chi_i = 1$ are $I = \{0, 2, 4\}$.

Using the formulas derived in the proof of Proposition 2.4 and properly accounting for set indices:

For $i = 0$, with $n = 0$: $|A_0| = i - n + 1 = 0 - 0 + 1 = 1$

For $i = 2$, with $n = 1$: $|A_1| = i - n + 1 = 2 - 1 + 1 = 2$

For $i = 4$, with $n = 2$: $|A_2| = i - n + 1 = 4 - 2 + 1 = 3$

Therefore, the collection of sets $A = \{A_0, A_1, A_2\}$ has the following cardinalities:

- $|A_0| = 1$
- $|A_1| = 2$
- $|A_2| = 3$

Following the Binary Decoder convention from Definition 2.3, we can determine both the index of the set and its content:

For $i = 0$, $\chi_0 = 1$:

- Using $i = n + |A_n| - 1$, we get $0 = n + |A_n| - 1$
- This gives $n = 0$, $|A_0| = 1$
- $A_0 = \{1\}$

For $i = 2$, $\chi_2 = 1$:

- Using $i = n + |A_n| - 1$, we get $2 = n + |A_n| - 1$
- This gives $n = 1$, $|A_1| = 2$
- $A_1 = \{1, 2\}$

For $i = 4$, $\chi_4 = 1$:

- Using $i = n + |A_n| - 1$, we get $4 = n + |A_n| - 1$
- This gives $n = 2$, $|A_2| = 3$
- $A_2 = \{1, 2, 3\}$

Therefore, we get the collection $\{\{1\}, \{1,2\}, \{1,2,3\}\}$.

Example 2.6 (Comparing Decoding Approaches).

Consider the binary string $b = 10101$ from Example 2.4. The indices where $b_i = 1$ are $I = \{0, 2, 4\}$.

Using the set cardinality recovery approach from Proposition 2.4:

$|A_0| = 0 + 1 = 1$

$|A_1| = 2 - 0 = 2$

$|A_2| = 4 - 2 = 2$

Following the Binary Decoder approach from Definition 2.3, we can determine both the index of the set and its content:

For $i = 0$, $b_0 = 1$:

- Solve $0 = n + |A_n| - 1$
- This gives $n = 1$, $|A_1| = 1$
- $A_1 = \{1\}$

For $i = 2$, $b_2 = 1$:

- Solve $2 = n + |A_n| - 1$
- This gives $n = 2$, $|A_2| = 1$
- $A_2 = \{1, 2\}$

For $i = 4$, $b_4 = 1$:

- Solve $4 = n + |A_n| - 1$
- This gives $n = 3$, $|A_3| = 2$
- $A_3 = \{1, 2, 3\}$

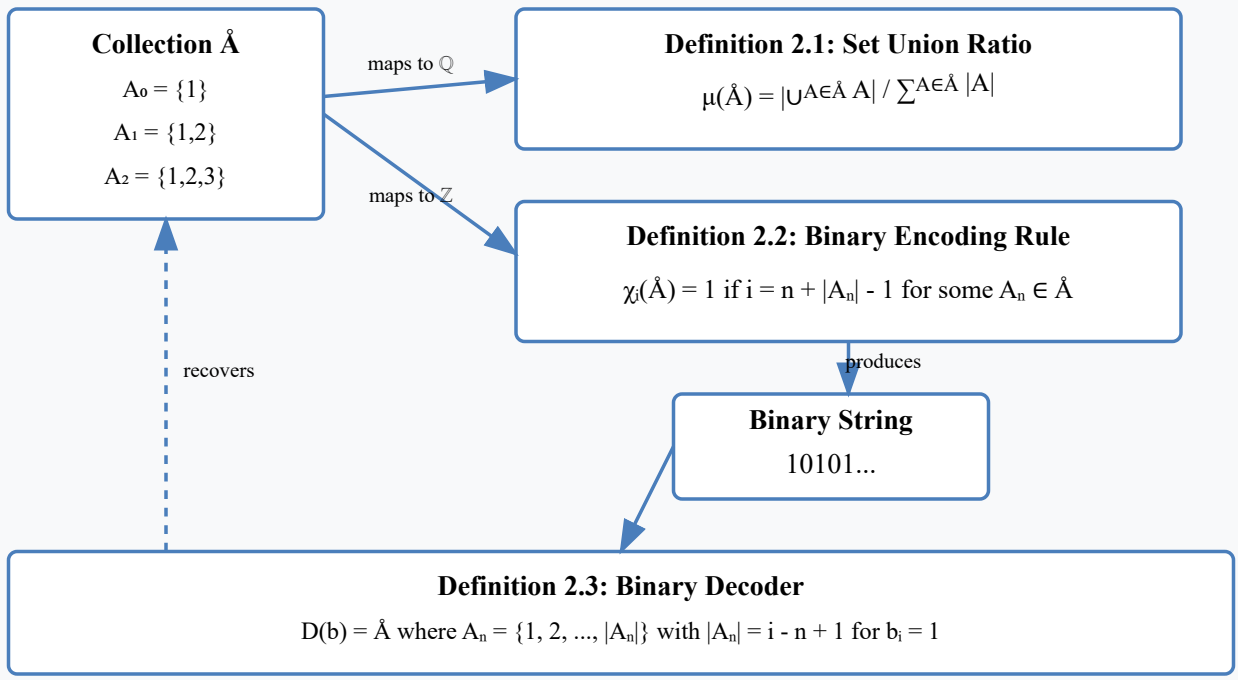
Therefore, when applying the Binary Decoder from Definition 2.3 to the binary string $b = 10101$, we get the collection $D(10101) = \{\{1\}, \{1,2\}, \{1,2,3\}\}$.

Note the difference in indexing between the two approaches:

- In Proposition 2.4, sets are indexed starting with A_0
- In Definition 2.3, sets are indexed starting with A_1

When properly accounting for this indexing difference, both approaches yield consistent results. The cumulative sizes of sets align with the positions of 1s in the binary string.

Fig. 1: Set-Theoretic Approach to Bijections



3. Constraints on Set Elements

Definition 3.1 (Index-Consistent Sets).

A collection of sets $A = \{A_0, A_1, ..., A_{m-1}\}$ is said to be *index-consistent* if for any two sets A_i and A_j in the collection, and for any index k, if both sets have an element at index k, then they have the same element at that index.

Example 3.1.

For the binary sequence $\chi = (1, 0, 1, 0, 0, 1)$ with set cardinalities $|A_0| = 1$, $|A_1| = 2$, and $|A_2| = 3$, an index-consistent collection might be:

Let $U = \{e_0, e_1, e_2, ...\}$ be a universe of elements. Then:

- $A_0 = \{e_0\}$
- $A_1 = \{e_0, e_1\}$
- $A_2 = \{e_0, e_1, e_2\}$

In this case, the collection forms a nested structure: $A_0 \subseteq A_1 \subseteq A_2$

The key property is that the element at position 0 in all three sets is e_0, and the element at position 1 in both A_1 and A_2 is e_1.

4. Proposed Bijection Approach

4.1 Mapping Strategy

The proposed approach for establishing a bijection between integers and rational numbers consists of:

1. Using the Binary Encoding Rule (χ_i) to establish a mapping between set collections and integers (via binary representation)
2. Using the Set Union Ratio (μ) to establish a mapping between the same set collections and rational numbers
3. Combining these mappings to create a bijection between integers and rational numbers

4.2 Modified Union Ratio for Extended Range

Definition 4.1 (Inverted Set Union Ratio).

The Inverted Set Union Ratio $\mu^{-1}: \mathcal{P}(\mathcal{P}(\mathbb{N})) \to [1,\infty)$ is defined as:

$$\mu^{-1}(\mathring{A}) = \sum_{A \in \mathring{A}} |A| \,/\, |\cup_{A \in \mathring{A}} A|$$

This inverted ratio allows representation of rational numbers ≥ 1.

TO DO: *Develop a complete extension of the Union Ratio to cover all positive rational numbers.*

5. Challenges and Limitations

5.1 Non-uniqueness Problem

The primary challenge to establishing a bijection is the non-uniqueness of the μ mapping. Different set collections can yield identical μ values, as demonstrated in the following theorem:

> **Theorem 5.1.**
>
> **There exist distinct collections of sets $\mathring{A} \neq \mathscr{B}$ such that $\mu(\mathring{A}) = \mu(\mathscr{B})$.**

Proof: Consider the following collections:

$\mathring{A} = \{\{1,2\}, \{1,2\}, \{1,2\}, \{1,2\}\}$

$\mathscr{B} = \{\{1,2\}, \{1\}, \{1\}, \{1\}, \{1\}, \{1\}, \{1\}\}$

For collection $\mathring{A}$:

$|\cup_{A\in\mathring{A}} A| = |\{1,2\}| = 2$

$\sum_{A\in\mathring{A}} |A| = 2 + 2 + 2 + 2 = 8$

Therefore, $\mu(\mathring{A}) = 2/8 = 0.25$

For collection $\mathscr{B}$:

$|\cup_{A\in\mathscr{B}} A| = |\{1,2\}| = 2$

$\sum_{A\in\mathscr{B}} |A| = 2 + 1 + 1 + 1 + 1 + 1 + 1 = 8$

Therefore, $\mu(\mathscr{B}) = 2/8 = 0.25$

Thus, $\mu(\mathring{A}) = \mu(\mathscr{B})$ despite $\mathring{A} \neq \mathscr{B}$.

TO DO: *Explore additional constraints on set collections to ensure uniqueness of the mapping.* □

5.2 Additional Requirements

For a valid bijection, the following conditions must be satisfied:

1. Injectivity: Different rational numbers must map to different binary encodings
2. Surjectivity: Every integer must correspond to some rational number
3. Constructibility: A systematic method for constructing the set collection for any rational number

TO DO: *Provide proofs for injectivity and surjectivity of the proposed mapping.*

6. Future Research Directions

Future research could explore:

- Generalizations of the approach to other number systems
- Connections to existing bijection proofs
- Development of algorithms for efficient computation of the mappings
- Properties of the binary sequences generated by specific types of set collections
- Applications of this mapping in coding theory or data structures
- Extensions to non-binary sequences or more complex set relationships

7. Conclusion

This paper has presented a novel set-theoretic approach to establishing a bijection between integers and rational numbers. We have established a bijection between collections of sets and certain binary sequences, demonstrating how to uniquely recover set cardinalities from binary representations. We have also explored constraints such as index consistency that add structure to the possible set collections.

While challenges remain, particularly regarding the uniqueness of the mapping due to the non-uniqueness problem of the